

Water as Simultaneous Fuel, Moderator and Reaction Medium for Proton–Electron Capture Chain Reactions with Lithium-6 Energy Extraction

Eymen Aydoğan

May 19, 2026

Abstract

We propose that ordinary water (H_2O) is an ideal reaction medium for the PECCNAL proton–electron capture mechanism, serving simultaneously as fuel source (hydrogen atoms), neutron moderator, and reaction containment. Each water molecule provides 10 electrons as potential capture targets (2 from hydrogen, 8 from oxygen), offering electron density 11,000 times higher than hydrogen gas at standard conditions. Produced neutrons are naturally moderated by hydrogen nuclei in water before reaching a dissolved Lithium-6 target, which releases 4.78 MeV per neutron capture. The system requires no cryogenic cooling, operates at room temperature, and uses the most abundant liquid on Earth as its primary fuel.

Keywords: water medium, neutron moderation, proton-electron capture, lithium-6, room temperature, LENR

1 Introduction

Previous papers in this series [?] employed hydrogen or deuterium gas as the reaction medium. Here we propose that liquid water offers substantial advantages as a reaction medium for the PECCNAL mechanism, combining high electron density, natural neutron moderation, and near-zero material cost.

2 Water as Reaction Medium

2.1 Electron Density Advantage

The electron density advantage of water means a free proton travelling through water encounters an electron target 111,000 times more frequently than in hydrogen gas at STP.

Table 1: Electron density comparison across reaction media

Medium	Density	e ⁻ /molecule	Rel. density
H ₂ gas (STP)	0.09 kg/m ³	2	1×
H ₂ gas (100 atm)	9 kg/m ³	2	100×
Liquid H ₂	70 kg/m ³	2	778×
Liquid H₂O	1000 kg/m³	10	111,000×

2.2 Natural Neutron Moderation

Produced neutrons must reach thermal energies for efficient Li-6 capture. In water:

$$n + {}^1\text{H} \rightarrow {}^2\text{H} + \gamma \quad \sigma_{\text{thermal}} = 0.33 \text{ barn} \quad (1)$$

Water is the most effective neutron moderator known, used in pressurized water reactors (PWR) worldwide. Neutrons thermalize within centimetres of their production point, maximizing Li-6 capture probability.

2.3 Lithium-6 Dissolution

Lithium carbonate (Li₂CO₃) dissolves in water, distributing Li-6 uniformly throughout the reaction volume:

$$n + {}^6\text{Li} \rightarrow {}^4\text{He} + {}^3\text{H} + 4.78 \text{ MeV} \quad (2)$$

3 Reaction Sequence in Water Medium

$$p + e_{(\text{H}_2\text{O})}^- \rightarrow n + \nu_e \quad - 1.293 \text{ MeV} \quad (\text{W1})$$

$$n + {}^1\text{H}_{(\text{H}_2\text{O})} \rightarrow {}^2\text{H} + \gamma \quad + 2.224 \text{ MeV} \quad (\text{W2})$$

$$n + {}^6\text{Li}_{(\text{dissolved})} \rightarrow {}^4\text{He} + {}^3\text{H} \quad + 4.780 \text{ MeV} \quad (\text{W3})$$

Net energy per cycle:

$$E_{\text{net}} = 2.224 + 4.780 - 1.293 = \boxed{+5.711 \text{ MeV}} \quad (3)$$

4 Practical Advantages

- **No cryogenics:** operates at room temperature
- **Self-shielding:** water absorbs most radiation
- **Abundant fuel:** seawater contains ~ 35 ppm Li
- **Low cost:** water and Li₂CO₃ are inexpensive
- **Scalable:** volume scales cubically with linear dimension

5 Conclusion

Water is proposed as the optimal reaction medium for PECCNAL, offering $111,000\times$ higher electron density than hydrogen gas, built-in neutron moderation, and compatibility with dissolved Li-6. Net energy output is 5.711 MeV per cycle with no cryogenic requirements.

Acknowledgments

The author thanks the nuclear and LENR physics communities.

References

- [1] E. Aydoğan, “Coulomb-Barrier-Free Neutron Production,” arXiv preprint, 2026.
- [2] R. L. Workman et al. (PDG), *Prog. Theor. Exp. Phys.*, 083C01, 2022.